

Embedded systems exp-16

Program:

```
#include <reg51.h>
#include <stdio.h>
void serial_ISR(void) interrupt 4
{
    char ch;
    if (RI)
    {
        RI = 0; // Clear receive interrupt flag
        ch = SBUF; // Read received character
        printf("\n>>> INTERRUPT RECEIVED: ");
        printf("%c\n", ch);
    }
}
void delay(void)
{
    unsigned int i;
    for(i = 0; i < 50000; i++); // Delay for visibility
}
void main(void)
{
    SCON = 0x50; // UART mode 1, 8-bit, REN enabled
    TMOD = 0x20; // Timer1 mode 2 (auto-reload)
    TH1 = 0xFD; // Baud rate 9600
    TR1 = 1; // Start Timer1
    IE = 0x90; // EA = 1, ES = 1 (enable serial interrupt)
    TI = 1; // Enable printf
    while(1)
```

```

{
printf("Hello World\n");
delay();
}
}

```

Output:

The screenshot shows the Keil uVision IDE with the following components:

- Registers:** A list of registers (r0-r7, Sys, a, b, sp, sp_max, PC, dptr, states, sec, psw) with their current values.
- Disassembly:** A window showing the assembly code for the main function, including the initialization of SCON and the call to printf and delay.
- Command:** A window showing the command line used to run the program: "Running with Code Size Limit: 2K Load 'C:\Keil_v5\C51\Examples\HELLO\keil\Objects\expl6'".
- UART #1:** A window showing the output of the program, which is "Hello World" printed four times.

When interrupt received

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- Registers:** A list of registers (r0-r7, Sys, a, b, sp, sp_max, PC, dptr, states, sec, psw) with their current values.
- Disassembly:** A window showing the assembly code for the main function, including the initialization of SCON and the call to printf and delay. It also shows the interrupt service routine (ISR) for interrupt 4, which reads a character from the serial port and prints it.
- Command:** A window showing the command line used to run the program: "Running with Code Size Limit: 2K Load 'C:\Keil_v5\C51\Examples\HELLO\keil\Objects\expl6'".
- UART #1:** A window showing the output of the program, which is "Hello World" printed four times, with an interrupt received during the delay. The output shows ">>> INTERRUPT RECEIVED: A" followed by "Hello World" printed four times.